COLLINS NJOKU

Mechatronics & Robotics Engineering Student

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EDUCATION

B.Eng. Mechatronics and Robotics Engineering

University :Belgorod State University

Expected Graduation: 2027

Relevant Coursework

- Artificial Intelligence for Robotics
- Computer Vision
- Control Systems
- Autonomous Systems
- Sensor Fusion

TECHNICAL SKILLS

Programming

Python, C++, MATLAB

Robotics

ROS / ROS2, SLAM, Motion Planning, Sensor Fusion, Gazebo

Computer Vision & AI

OpenCV, YOLO, PyTorch, TensorFlow, Deep Learning

Tools

Linux, Git, Docker

Companies hiring robotics engineers typically expect **strong programming skills (Python/C++), ROS, computer vision libraries, and experience with sensors or perception algorithms**.

PROJECTS

Autonomous Navigation with Object Detection

Python, OpenCV, YOLO

- Built real-time perception pipeline for autonomous robot navigation
- Implemented YOLO-based object detection for obstacle recognition
- Integrated computer vision with robot decision-making logic
- Achieved real-time inference using optimized image processing pipeline

Visual Servoing Robotic Arm (Simulation)

Python, Robotics Control

- Designed closed-loop visual control system for robotic arm manipulation
- Implemented target tracking using camera feedback
- Reduced positional error using vision-based control algorithms

AI Voice Controlled Assistant

Python, SpeechRecognition, NLP

- Developed voice-based command system with authentication
- Implemented wake-word detection and command recognition
- Integrated secure voice password system

RESEARCH INTERESTS

Robotics Perception

Autonomous Vehicles

Robotic Manipulation

Computer Vision for Robotics

KEY ACHIEVEMENTS

- Developed multiple robotics perception projects using deep learning
- Implemented real-time computer vision pipelines
- Built robotics simulation systems combining perception and control